When does the default PING circuit start to sing?

exp083 · 17 August 2026 · Draft

Abstract

The default `snn.components.ping()` circuit is silent at weak drive, begins firing near 50 Hz per input channel, and settles into reproducible gamma activity from 75 to 200 Hz. Across that resolved interval, the median gamma frequency rises from 37.85 to 56.19 Hz. One compiled graph produces the entire response curve. No model parameter is tuned between conditions.

Methods

1. **Author one graph.** A typed 128-channel spike input drives the default PING component, which contains 80 excitatory (E) and 20 inhibitory (I) cells. The timestep is 0.1 ms. Neuron, synapse, delay, initializer, and constraint specifications remain at their component defaults.
2. **Vary only input drive.** Homogeneous Poisson input spans 0 Hz, 25 Hz, 50 Hz, 75 Hz, 100 Hz, 125 Hz, 150 Hz, 200 Hz per channel. Every condition uses the same compiled graph and 5 paired deterministic trials, each lasting 1000 ms. Measurements exclude the first 200 ms.
3. **Resolve gamma consistently.** The `default-ping-gamma-v1` policy computes a single-trial Welch spectrum from the E-population raster. It searches 30–80 Hz, interpolates the peak between bins, rejects maxima at the band edge, and requires peak power to exceed 3 times the band median. Failed checks retain an unresolved reason; they are not reported as 0 Hz.
4. **Score rhythmicity.** The standard exp054 metric bins the E-population spike count at 1 ms, computes its normalized autocorrelogram, and applies a three-point smoothing kernel. If L is the first side-lobe height and T the following trough height, the dimensionless Michelson contrast is $R = \frac{L-T}{L+T}$. Silence is reported as $R = 0$.
5. **Measure E/I timing separately.** Post-transient population spikes are binned at 1 ms. The lag of the strongest E/I cross-correlation within plus or minus 20 ms is descriptive and does not enter the gamma-resolution rule.

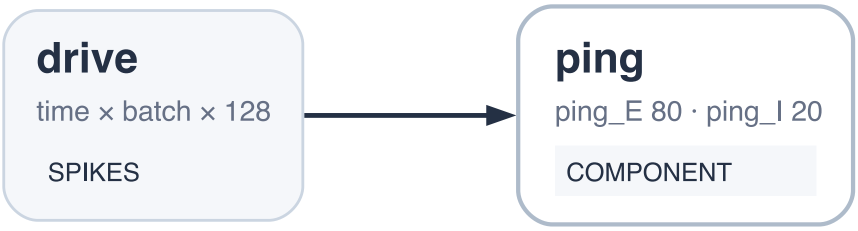

Figure 1: Compiled graph reused at every input condition. A time-by-batch-by-128 spike tensor drives one default PING component with 80 E cells and 20 I cells. Only the input event probability changes across the sweep; the graph and its parameters remain fixed.

Results

The network is silent through 25 Hz per channel. At 50 Hz it activates and the median rhythmicity score reaches 1, but gamma resolves in only 20% of trials. From 75 Hz onward, every trial resolves. The contrast detects strong temporal structure immediately at onset, whereas the spectral rule separately asks whether a stable peak lies inside the registered gamma band.

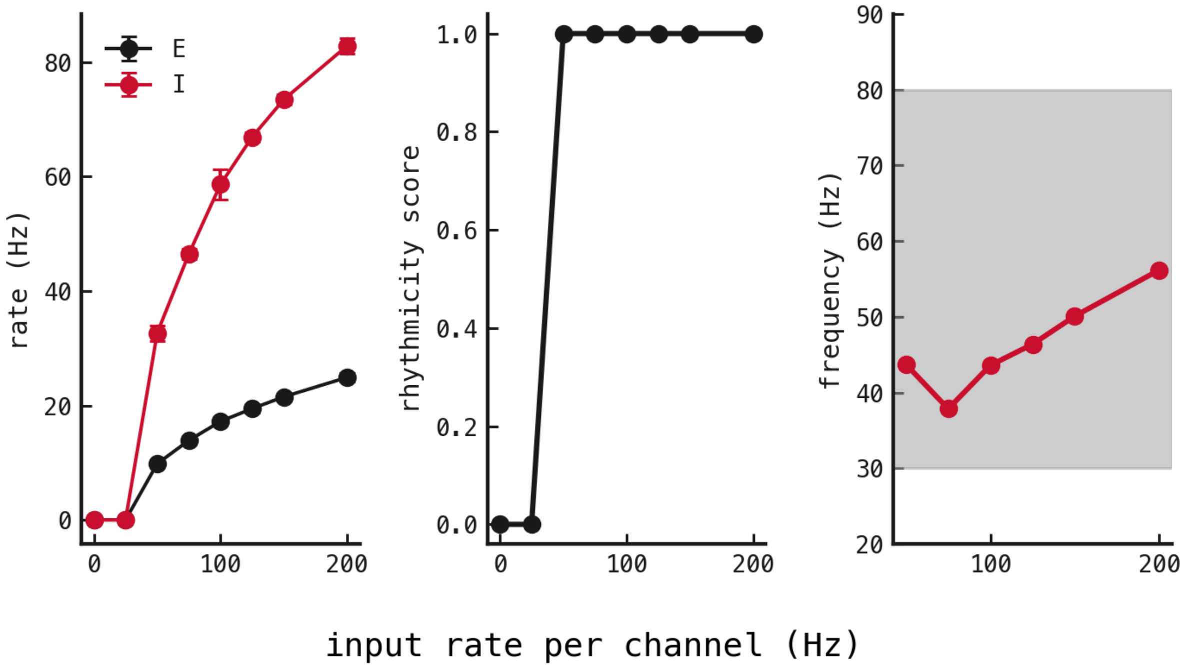

Figure 2: Response of one fixed default PING graph to homogeneous Poisson drive. Left: mean E and I firing rates in hertz, with standard deviations across 5 paired trials. Middle: median lobe-trough rhythmicity contrast R, with error bars showing half the interquartile range; $R = 0$ is flat and R approaches 1 as periodic structure strengthens. Right: median resolved E-population frequency in hertz; the grey region marks the registered 30–80 Hz search band. At 50 Hz per channel, R already saturates while the spectral rule resolves only 20% of trials, separating temporal structure from gamma-band peak validity.

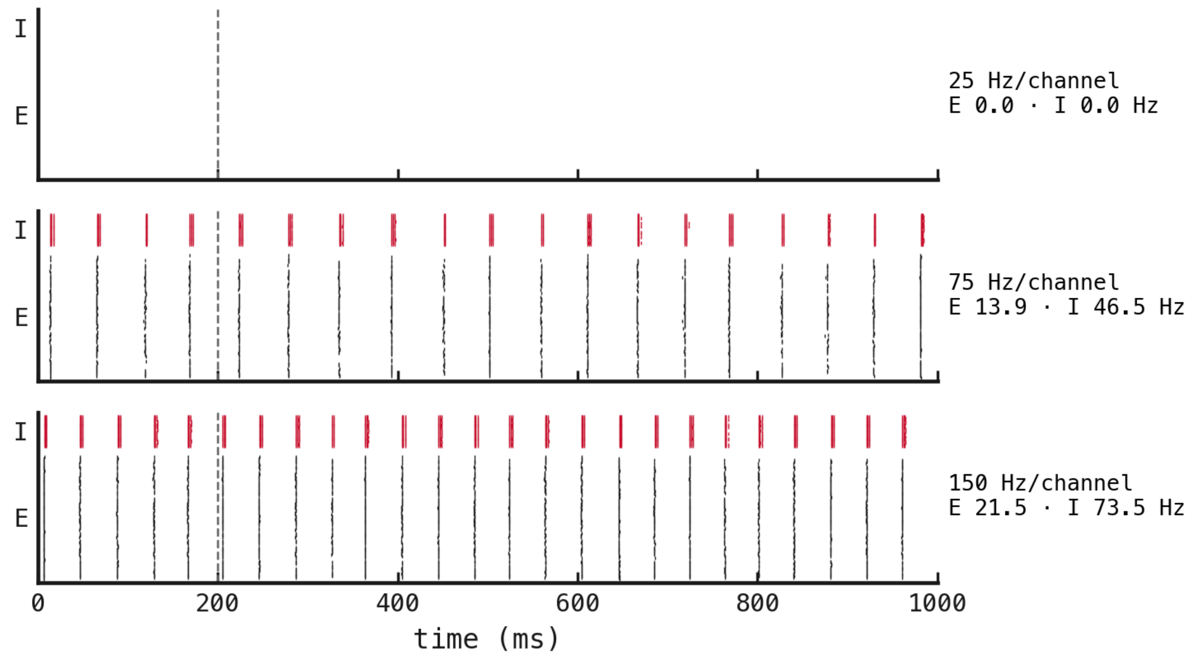

Figure 3: Single-trial E and I rasters at the three input rates fixed before the primary sweep. Time is in milliseconds; E cells are black, I cells are red, and the dashed vertical line marks the 200 ms transient exclusion. The right labels report input rate per channel and post-transient mean population rates in hertz. The 25 Hz condition is silent, whereas the higher-drive conditions show repeated population volleys.

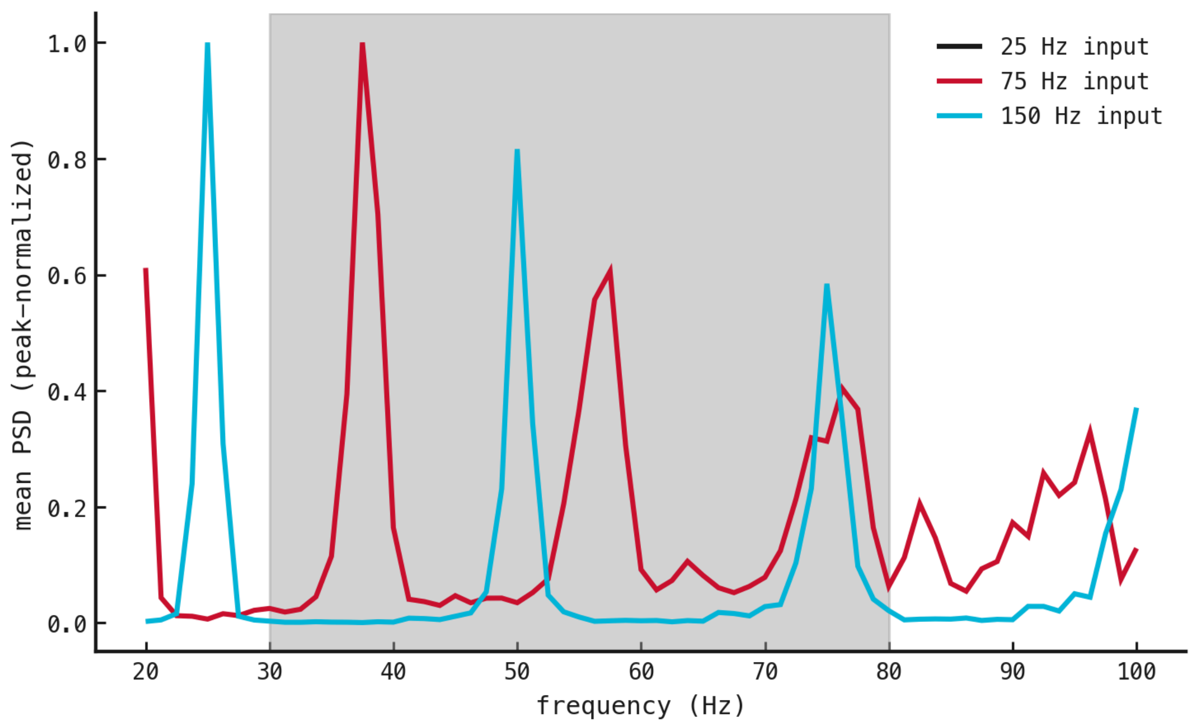

Figure 4: Mean E-population power spectra at the preselected input rates. Frequency is in hertz and power is normalized to each curve's maximum for shape comparison; gamma resolution uses the unnormalized single-trial spectra. The grey region marks 30–80 Hz. Stronger drive shifts the dominant resolved rhythm upward within the registered band.